

University of Cape Town

Bayesian Computational Methods

Assignment

Help

Author:
Isabella Lethbridge
Tayyeb Abrahams

Student Number:
LTHISA001
ABRMOE030

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Introduction

Question a

Each of the n birds is caught exactly once. The capture day of bird i is denoted as t_i and is treated as fixed. Every birds moults at the same constant rate, with moult lasting τ days for all birds. The start of moult for bird i, T_i, is an unobserved random variable, where

$$T_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2), \qquad f_T(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right), \qquad i = 1, \dots, n.$$

Let $Z = \frac{T_i - \mu}{\sigma} \sim \mathrm{N}(0,1)$ and let Φ denote the corresponding CDF. In the Type 1 model, the moult scores (y_i) are not observed, rather the status $(s_i \in \{1,2,3\})$ of each bird is observed.

For $s_i = 3$, bird i has not yet started moult:

$$P(t_i) = \Pr(T_i > t_i) = \Pr\left(Z > \frac{t_i - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{t_i - \mu}{\sigma}\right).$$

For $s_i = 2$, bird i is currently moulting:

$$Q(t_i) = \Pr(t_i - \tau \le T_i \le t_i) = \Pr\left(\frac{t_i - \tau - \mu}{\sigma} \le Z \le \frac{t_i - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \mu}{\sigma}\right) - \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right).$$

For $s_i = 1$, bird i has finished moulting:

$$R(t_i) = \Pr(T_i \le t_i - \tau) = \Pr\left(Z \le \frac{t_i - \tau - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right).$$

Given t_i, the status s_i is a single categorical draw with probabilities $P(t_i), Q(t_i)$ and $R(t_i)$. Thus, we can write the conditional probability of s_i as

$$\Pr(s_i \mid t_i, \mu, \sigma^2, \tau) = P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}}.$$

Since the birds are independent, the likelihood can be written as the product over all n birds:

$$\begin{aligned} L(\boldsymbol{s}, \boldsymbol{t}, \mu, \sigma^2, \tau) &= \prod_{i=1}^{n} P(t_i)^{\mathrm{I}\{s_i=3\}}\, Q(t_i)^{\mathrm{I}\{s_i=2\}}\, R(t_i)^{\mathrm{I}\{s_i=1\}} \\ &= \prod_{i=1}^{n} \left[1 - \Phi(\frac{t_i - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=3\}} \left[\Phi(\frac{t_i - \mu}{\sigma}) - \Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=2\}} \left[\Phi(\frac{t_i - \tau - \mu}{\sigma})\right]^{\mathrm{I}\{s_i=1\}}. \end{aligned}$$

On any capture day t, the population consists of birds which belong to three different states. Birds have either not yet started to moult, are undergoing moult or have completed moult. These states occur in proportions $P(t), Q(t)$ and $R(t)$, which depend on t through μ, σ^2 and τ. Thus, the moult index has a mixture distribution with a point mass at 0 and 1 with weights $P(t)$ and $R(t)$, respectively, as well as a continuous component on $(0,1)$ with weight

$Q(t)$. In the Type 1 model, the moult index is not observed. We only observe the state that bird i belongs to. Thus, each bird contributes the weight of the state it belongs to $P(t_i)$, $Q(t_i)$ or $R(t_i)$, to the likelihood.

Question b

In the Type 2 model, the moult scores (y_i) for birds that are currently moulting are observed. We group the birds according to their status. Let $\mathcal{F} = \{i : s_i = 1\}$ denote the set of birds that have already moulted, $\mathcal{M} = \{i : s_i = 2\}$ denote set of birds that are undergoing moult and $\mathcal{N} = \{i : s_i = 3\}$ denote the set of birds that have not yet started to moult. Let $m = |\mathcal{M}|$ be the number of birds currently moulting.

For a bird undergoing moult, the linear assumption gives $T_i = t_i - y_i\tau$. The probability density function of its observed moult score is

$$f_y(y_i) = \tau f_T(t_i - y_i\tau) \qquad 0 < y < 1.$$

The likelihood contribution for birds that have already moulted and birds that have not yet started to moult is the same as in the Type 1 model. The Type 2 likelihood is therefore

$$L(\boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \mu, \sigma^2, \tau) = \prod_{i\in\mathcal{F}} R(t_i) \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} P(t_i).$$

The joint posterior distribution is

$$\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto \prod_{i\in\mathcal{F}} R(t_i) \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} P(t_i)\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau),$$

with the following prior distributions $\mu \sim \text{Uniform}(100, 160)$, $\pi(\sigma^2) \sim \text{IG}(1, 1)$, $\tau \sim \text{Uniform}(50, 180)$.

Data augmentation

The probabilities $R(t_i)$ and $P(t_i)$ are integrals over the unobserved start date

$$R(t_i) = \Pr(T_i \le t_i - \tau) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}\, dT_i$$
$$P(t_i) = \Pr(T_i > t_i) = \int_{-\infty}^{\infty} f_T(T_i)\, \mathrm{I}\{T_i > t_i\}\, dT_i.$$

The status is fully determined by the start date (T_i), so each integrand is the joint density of the start date and the bird's status

$$p(T_i, s_i = 1 \mid \mu, \sigma^2, \tau) = f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\},$$
$$p(T_i, s_i = 3 \mid \mu, \sigma^2) = f_T(T_i)\, \mathrm{I}\{T_i > t_i\}.$$

We treat T_i for $i \in \mathcal{F} \cup \mathcal{N}$ as latent variables and sample from the augmented posterior

$$\pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto \prod_{i\in\mathcal{F}} \left[f_T(T_i)\, \mathrm{I}\{T_i \le t_i - \tau\}\right] \prod_{i\in\mathcal{M}} \left[\tau f_T(t_i - y_i\tau)\right] \times \prod_{i\in\mathcal{N}} \left[f_T(T_i)\, \mathrm{I}\{T_i > t_i\}\right] \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau).$$

Integrating out each T_i returns $R(t_i)$ or $P(t_i)$

$$\int \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})\, d\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})} = \pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}).$$

The posterior distribution of interest is the marginal of the augmented posterior. The draws of (μ, σ^2, τ) from a sampler targeting the augmented posterior are draws from the required posterior. No latent variables are needed for the moulting birds, since $T_i = t_i - y_i\tau$ is deterministic for $i \in \mathcal{M}$. Since $0 < y_i < 1$, $t_i - \tau \le T_i \le t_i$ is always satisfied.

Conditional distributions of T_i

The conditional distributions of the latent start dates are the same truncated normal distributions as in the Type 1 Gibbs sampler:

$$\begin{aligned} T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i \le t_i - \tau\}, \quad i \in \mathcal{F}, \\ T_i \mid \cdot &\sim \mathrm{N}(\mu, \sigma^2)\, \mathrm{I}\{T_i > t_i\}, \qquad i \in \mathcal{N}. \end{aligned}$$

Conditional distribution of μ

Keeping only the factors that involve μ, namely the n normal densities and the prior,

$$\begin{aligned} \pi(\mu \mid \cdot) &\propto \prod_{i\in\mathcal{F}} \left[f_T(T_i)\right] \prod_{i\in\mathcal{M}} \left[f_T(t_i - y_i\tau)\right] \prod_{i\in\mathcal{N}} \left[f_T(T_i)\right] \pi(\mu) \\ &\propto \prod_{i\in\mathcal{F}} \left[f_T(T_i)\right] \prod_{i\in\mathcal{M}} \left[f_T(T_i)\right] \prod_{i\in\mathcal{N}} \left[f_T(T_i)\right] \pi(\mu) \\ &\propto \prod_{i=1}^{n} f_T(T_i)\, \pi(\mu) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(T_i-\mu)^2}{2\sigma^2}\right) \mathrm{I}\{100 \le \mu \le 160\} \\ &\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(T_i-\mu)^2\right) \mathrm{I}\{100 \le \mu \le 160\}. \end{aligned}$$

Let $\bar{T} = \frac{1}{n}\sum_{i=1}^{n} T_i$.

$$\begin{aligned}\sum_{i=1}^{n}(T_i-\mu)^2 &= \sum_{i=1}^{n}\left[(T_i-\bar{T})+(\bar{T}-\mu)\right]^2 \\ &= \sum_{i=1}^{n}(T_i-\bar{T})^2 + 2(\bar{T}-\mu)\sum_{i=1}^{n}(T_i-\bar{T}) + n(\bar{T}-\mu)^2 \\ &= \sum_{i=1}^{n}(T_i-\bar{T})^2 + n(\bar{T}-\mu)^2,\end{aligned}$$

since $\sum_{i=1}^{n}(T_i-\bar{T})=0$.

The first term does not depend on μ, so

$$\pi(\mu \mid \cdot) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{T})^2\right) \mathrm{I}\{100 \le \mu \le 160\}.$$

This is the kernel of a truncated normal distribution

$$\mu \mid \cdot \sim \mathrm{N}\left(\bar{T}, \frac{\sigma^2}{n}\right) \mathrm{I}\{100 \le \mu \le 160\}.$$

For the moulting birds, $T_i = t_i - y_i\tau$ is used in $\bar{T}$.

Conditional distribution of σ^2

Keeping only the factors that involve σ^2, namely the n normal densities and the prior, Let $S=\sum_{i=1}^{n}(T_i-\mu)^2$,

$$\begin{aligned}\pi(\sigma^2 \mid \cdot) &\propto \prod_{i\in\mathcal{F}}\left[f_T(T_i)\right] \prod_{i\in\mathcal{M}}\left[f_T(t_i-y_i\tau)\right] \prod_{i\in\mathcal{N}}\left[f_T(T_i)\right]\, \pi(\sigma^2) \\ &\propto \prod_{i=1}^{n} f_T(T_i)\, \pi(\sigma^2) \\ &\propto (\sigma^2)^{-n/2}\exp\left(-\frac{S}{2\sigma^2}\right)(\sigma^2)^{-2}\exp\left(-\frac{1}{\sigma^2}\right) \\ &= (\sigma^2)^{-(\frac{n}{2}+1)-1}\exp\left(-\frac{1+\frac{S}{2}}{\sigma^2}\right).\end{aligned}$$

This the kernel of an inverse-gamma distribution

$$\sigma^2 \mid \cdot \sim \mathrm{IG}\left(\frac{n}{2}+1,\ 1+\frac{1}{2}\sum_{i=1}^{n}(T_i-\mu)^2\right).$$

This differs from the Type 1 conditional given in the assignment, which corresponds to the prior $\pi(\sigma^2) \propto 1/\sigma^2$ rather than $\mathrm{IG}(1,1)$.

Conditional distribution of τ

Keeping only the factors that involve τ, namely the densities of the moulting birds (with $T_i = t_i - y_i\tau$), the indicators of the finished birds and the prior,

$$\begin{aligned}\pi(\tau \mid \cdot) &\propto \prod_{i\in\mathcal{F}} \left[\mathrm{I}\{T_i \le t_i - \tau\}\right] \prod_{i\in\mathcal{M}} \left[\tau f_T(t_i - y_i\tau)\right] \pi(\tau) \\ &\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\}\, \mathrm{I}\{50 \le \tau \le 180\}.\end{aligned}$$

Moulting birds.

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\frac{1}{2\sigma^2}\sum_{i\in\mathcal{M}} (t_i - \mu - y_i\tau)^2\right).$$

Expanding the square,

$$\sum_{i\in\mathcal{M}} (t_i - \mu - y_i\tau)^2 = \sum_{i\in\mathcal{M}} (t_i - \mu)^2 - 2\tau \sum_{i\in\mathcal{M}} y_i(t_i - \mu) + \tau^2 \sum_{i\in\mathcal{M}} y_i^2,$$

where the first term does not depend on τ.

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right),$$

where $\beta = \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2}$ and $\gamma = \frac{\sum_{i\in\mathcal{M}} y_i(t_i - \mu)}{\sigma^2}$.

Finished birds.

The product of the indicators equals one only if this holds for every finished bird, so

$$\prod_{i\in\mathcal{F}} \mathrm{I}\{T_i \le t_i - \tau\} = \mathrm{I}\{\tau \le \min_{i\in\mathcal{F}}(t_i - T_i)\}$$

Let $h = \min_{i\in\mathcal{F}}(t_i - T_i)$.

That is, the duration of moult cannot exceed the time between the start date and the capture date of any finished bird.

Combining.

$$\pi(\tau \mid \cdot) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right) \mathrm{I}\{50 \le \tau \le \min(180, h)\}.$$

This is the kernel of a Modified-Half-Normal distribution (Sun et al., 2021)

$$\tau \mid \cdot \sim \text{MHN}\left(m+1,\ \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2},\ \frac{\sum_{i\in\mathcal{M}} y_i(t_i-\mu)}{\sigma^2}\right) \text{I}\{50 \le \tau \le \min(180, h)\}.$$

In the Type 1 model, τ appears only in the indicators and has a uniform conditional distribution. In the Type 2 model, the observed moult scores result in a non-uniformly shaped conditional of τ. Sampling from this distribution can be simplified using a further data augmentation step (Lemma 5 of Sun et al., 2021). The form of this data augmentation depends on the sign of γ. Since $y_i > 0$, $\gamma \ge 0$ if and only if $\mu \le \bar{t}_y = \sum_{i\in\mathcal{M}} y_i t_i \big/ \sum_{i\in\mathcal{M}} y_i$.

Case $\gamma \ge 0$. Introduce a latent variable V such that $V \mid \tau \sim \text{Poisson}(\gamma\tau)$. Multiplying the conditional of τ by the Poisson probability gives

$$\tau^m e^{-\beta\tau^2+\gamma\tau} \cdot \frac{e^{-\gamma\tau}(\gamma\tau)^V}{V!} \propto \tau^{m+V} e^{-\beta\tau^2},$$

so the term $e^{\gamma\tau}$ cancels. Substituting $W = \tau^2$ and noting that the indicator only involves τ,

$$W \mid V \sim \text{Gamma}\left(\frac{m+1+V}{2},\ \beta\right) \text{I}\{50^2 \le W \le \min(180, h)^2\},$$

where β is the rate parameter. When $\gamma = 0$, $V = 0$ and this reduces to a truncated $\text{Gamma}\left(\frac{m+1}{2}, \beta\right)$ distribution.

Case $\gamma < 0$. Using the parameterisation $\text{GIG}(p, a, b)$ with density proportional to $u^{p-1}\exp\{-(au + b/u)/2\}$, introduce a latent variable U such that

$$\begin{aligned} U \mid \tau &\sim \text{GIG}\left(\frac{1}{2},\ 1,\ \gamma^2\tau^2\right), \\ W \mid U &\sim \text{Gamma}\left(\frac{m+1}{2},\ \beta + \frac{\gamma^2}{2U}\right) \text{I}\{50^2 \le W \le \min(180, h)^2\}. \end{aligned}$$

In both cases, the truncated gamma distribution can be sampled exactly using the inverse CDF method, after which $\tau = \sqrt{W}$. The latent variable V or U is redrawn given the current τ at every iteration and then discarded, so the conditionals of the other parameters are unchanged.

The Gibbs sampling algorithm

1. Initialise $\mu^{(0)} \sim \text{U}(100, 160)$, $\sigma^{2(0)} \sim \text{IG}(1, 1)$ and $\tau^{(0)} \sim \text{U}(50, 180)$. Draw $T_i^{(0)}$ for $i \in \mathcal{F} \cup \mathcal{N}$ from their truncated normal conditionals given these values.
2. For $k = 1, \ldots, K$:
   a. For $i \in \mathcal{F}$, draw $T_i^{(k)} \sim \text{N}\big(\mu^{(k-1)}, \sigma^{2(k-1)}\big)\, \text{I}\{T_i \le t_i - \tau^{(k-1)}\}$.
   b. For $i \in \mathcal{N}$, draw $T_i^{(k)} \sim \text{N}\big(\mu^{(k-1)}, \sigma^{2(k-1)}\big)\, \text{I}\{T_i > t_i\}$.
   c. For $i \in \mathcal{M}$, set $T_i^{(k)} = t_i - y_i\tau^{(k-1)}$.
   d. Draw $\mu^{(k)} \sim \text{N}\big(\bar{T}^{(k)}, \sigma^{2(k-1)}/n\big)\, \text{I}\{100 \le \mu \le 160\}$.

e. Draw $\sigma^{2(k)} \sim \text{IG}(\frac{n}{2}+1,\ 1+\frac{1}{2}\sum_{i=1}^{n}(T_i^{(k)}-\mu^{(k)})^2)$.

f. Compute $h^{(k)} = \min_{i\in\mathcal{F}}(t_i - T_i^{(k)})$, and $\beta^{(k)}$ and $\gamma^{(k)}$ evaluated at $\mu^{(k)}$ and $\sigma^{2(k)}$. If $\gamma^{(k)} \geq 0$, draw $V \sim \text{Poisson}(\gamma^{(k)}\tau^{(k-1)})$ and $W \sim \text{Gamma}(\frac{m+1+V}{2}, \beta^{(k)})$; otherwise, draw $U \sim \text{GIG}(\frac{1}{2}, 1, (\gamma^{(k)}\tau^{(k-1)})^2)$ and $W \sim \text{Gamma}(\frac{m+1}{2}, \beta^{(k)} + \frac{(\gamma^{(k)})^2}{2U})$. In both cases, W is truncated to $[50^2, \min(180, h^{(k)})^2]$. Set $\tau^{(k)} = \sqrt{W}$.

3. After discarding a burn-in period, retain the draws $(\mu^{(k)}, \sigma^{2(k)}, \tau^{(k)})$ as samples from the posterior $\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})$. The latent variables $T_i^{(k)}$, V and U are discarded.

Question c

We sample from the likelihood surface using a Metropolis Hastings algorithm. However we have the problem that the steps of our random walk are drawn a multivariate normal distribution. But each of our three parameters do not have support over the entire real line: μ, which determines the average start time, is constrained by a uniform prior to be situated between 100–160; τ, which determines the speed of moulting, is also contrainted by its prior to be situated within 50–180; finally the variance is strictly positive. We define the following transformations:

$$\begin{aligned} f(\mu) &= \text{logit}\frac{\mu - 100}{60} \\ g(\tau) &= \text{logit}\frac{\tau - 50}{130} \\ h(\sigma^2) &= \log \sigma^2 \end{aligned}$$

These transformation each have support over the entire real line; thus we may perform our random walk over this transformed space. Since we are unsure how our parameters are correlated; to begin with, the steps of our random walk are drawn from:

$$e_i \sim \mathcal{N}(0, c\, I)$$

For some constant c. However, when back transform (let θ be our original parameters and ϕ their transformations) we must include the jacobian term in the posterior likelihood in order to ensure that the Metropolis-Hasting algorithm converges to the appropriate likelihood; the log-likelihood becomes:

$$L(\theta) + \log(\pi(\theta)) + \log|J(\phi)|$$

Where $J(\phi)$ is the Jacobian of our transformation. However, we would like to tune our random walk to have a more appropriate covaraince matrix; to achieve this we utilize 3000 burnin iterations and estimate the covariance matrix Σ of our transformed parameters on these iterations. Them each of our retained draws each have their random walk drawn from:

$$e \sim \mathcal{N}(0, \Sigma)$$

And the Metropolis Hastings algorithm algorithm proceeds as per usual. We obtain the following estimates of our parameters and as well as 95% credible intervals on them:

Table 1: Posterior mean estimates and 95% credible intervals for the Type 3 moulting model.

Parameter	Posterior mean	2.5% quantile	97.5% quantile
Mean start time (μ)	130.67	114.61	152.40
Variance ($\sigma 2$)	317.18	195.19	525.00
Moulting duration (τ)	92.79	56.43	128.40

We